Blue Ridge FloodWatch — Technical Section (slide-by-slide content)

Pulled from your `analysis.html` data-pipeline report and the modeling work in this chat. Each block below = one slide: a suggested title, the talking points, and which figure to drop in. Figures are in `eda_figures.zip` (18 charts pulled straight out of your R analysis doc) plus the 3 new ones (`pr_curve.png`, `calibration_plot.png`, `shap_summary.png`).

PART A — Where the data came from

Slide: The core question

- Which storm events in North Carolina become **high-consequence** (deadly or very costly), and what features of the landscape and built environment around them — steep terrain, nearby streams, road crossings, paved surfaces — help explain it?
- Four-step pipeline: NOAA storm records → landscape/infrastructure layers → spatial features measured around each event point → ML-ready table → model.
- **Figure:** none needed, or redraw the flowchart from section 1.3 (NOAA → event table; landscape data → spatial features; both → ML-ready data → model).

Slide: Primary data source — NOAA Storm Events

- Source: NOAA NCEI Storm Events Database, bulk CSVs, 2016–2026 (33 files), filtered to North Carolina. 2026 is a partial year (latest record June 30, 2026).
- Three linked tables per year: **details** (event type, place, time, deaths, injuries, damage, narrative), **fatalities** (one row per person killed), **locations** (one-plus points per event).
- Why NOAA: it's the official record of which storms hurt people/property — it defines the outcome you're predicting.
- Caveat to state on the slide: damage values are nominal-dollar estimates; many forecast-zone records have no coordinates.

Slide: Why NOAA alone isn't enough

Three gaps, each mapped to a dataset you added:

Gap	Dataset added	What it measures
Outcomes, not conditions (NOAA says *what happened*, not *why* one crossing flooded and another didn't)	—	—
Only ~1% of events cause a death — need place-based context to separate routine vs. deadly	—	—
27% of events have no coordinates (but 98% of flood-related events do)	—	—

Question	Dataset	Measures
Near a stream/river?	USGS NHD High Resolution + WBD watersheds	distance to nearest stream, stream density, containing HUC12
Steep/rugged ground?	USGS 3DEP-derived elevation/slope/roughness	elevation, slope, roughness at point + 250m buffer
Road crossing water nearby?	NCDOT bridges/structures/culverts	distance to + count of nearby crossings
Paved vs. natural?	USGS Annual NLCD imperviousness	% impervious at point + 250m/1km buffers
Soil absorbency?	USDA SSURGO	statewide soil summary (not yet event-mapped)

- **Figure:** `fig09_terrain_elevation_slope.png` or `fig11_nlcd_imperviousness_map.png` as a visual anchor.

Slide: The five supporting datasets (source credibility slide)

One line each — good for a "we didn't just use one dataset" credibility slide:

- **USGS NHD High Resolution** — stream/river network + watershed boundaries (WBD).
- **USGS 3DEP** (via elevatr/AWS Terrain Tiles) — elevation, slope, terrain roughness.
- **NCDOT Structures service** — inventoried bridges, culverts, road crossings.
- **USGS Annual NLCD** — fractional impervious-surface layer (2023 vintage).
- **USDA SSURGO** — soil survey summaries.

- Two candidate datasets **not yet included** (say this out loud — shows honesty/roadmap): rainfall totals (Stage IV/gridMET) and community vulnerability (SVI) — both scaffolded (`include_rainfall_features`, `include_community_features`) but not wired in yet at analysis time. *(Note: SOVI/population data was pulled in during the modeling stage of this project from FEMA's National Risk Index — mention that as the fix.)*
- **Figures:** `fig07_nhd_streams_map.png`, `fig08_wbd_huc12_watersheds.png`, `fig10_ncdot_road_crossings.png`, `fig12_ssurgo_soils_summary.png` — pick 2–3, don't use all 4.

PART B — What the data actually shows (numbers + EDA)

Slide: Scale of the raw data

- NOAA: 2016–2026, North Carolina only, collapsed from 3 linked tables to **one row per event** (naive joins would double-count deaths/injuries/damage across multiple fatality/location rows — worth a sentence: "we collapse with max(), never sum(), to avoid inflating impact totals").
- Region assignment (county vs. forecast-zone records) — **every** record matched to a region; zero fell through to "unmatched":

Method	Central	Coastal	Mountains
exact county match	6,710	5,064	3,032
forecast-zone → county	0	475	521
special coastal zone	0	87	0

- **Figure:** `fig01_region_map.png` (three-region scheme + 11-county Western NC study area, outlined).

Slide: Storm activity over time & seasonality

- Events per year across NC, by region — shows overall volume and the partial 2026 year.
- Flood-related events by month/year heatmap: clearly seasonal, Aug–Sep–Oct (hurricane season) and summer thunderstorm months dominate — this is the empirical basis for the `peak_hurricane_season` engineered feature used later.
- **Figures:** `fig02_events_by_year_region.png`, `fig03_flood_calendar_heatmap.png`.

Slide: Which storms cause harm

- Top event types by region, and which event types carry the most reported damage / deaths / injuries statewide.
- **Figures:** `fig04_top_event_types_by_region.png`, `fig05_impacts_by_event_type.png`.

Slide: The Helene story (your hook slide — use early or here)

- In the 11 study counties, **one week (Helene, late Sept 2024) accounts for 85% of storm deaths and 88% of reported damage across 2016–2026.**
- "In the mountains, a rare event can outweigh a decade of ordinary storms" — this is your motivating quote/thesis line.
- **Figure:** `fig06_study_area_damage_timeline.png` (monthly damage bar chart with the Helene spike called out).

Slide: Data quality / honesty slide

- 27% of all events lack coordinates (mostly forecast-zone records); 98% of flood-related events DO have coordinates, so the study isn't crippled by this.
- Missingness is never filled with zero — it's left as missing and flagged.
- **Figure:** `fig13_predictor_missingness.png` (share of events missing each predictor).

PART C — Building the feature table

Slide: How features are measured around each event

- Point-and-buffer methodology: nearest-line/point distance, counts within 250m/1km circular buffers, and mean raster values within a buffer.
- 11,610 events got spatial features computed. Applies to point measurements only — **no upstream contributing area / flow accumulation / HAND** is computed (say this explicitly, it's in your own limitations list and shows rigor).

Feature group	Source	Examples
Stream proximity	NHD HR	distance_to_stream_m, streams_within_250m/1km, stream density
Watershed	WBD	huc12, huc12_area_km2
Road crossings	NCDOT	distance/count of nearby culverts & structures
Terrain	3DEP	elevation_m, slope_degrees, terrain_roughness (+250m means)
Imperviousness	NLCD	impervious_mean_event/250m/1km

- **Figure:** table above, or `fig18_study_area_streams_crossings_map.png` as a visual.

Slide: Outcome label & leakage discipline

- Target: `high_consequence` = 1 if a death occurred, **or** damage exceeded the 90th percentile of positive damage values ($100,000 basis on the full dataset — **recomputed on training rows only** before final evaluation to avoid leakage). 650 positives, 4.1% prevalence.
- Secondary label (reported, not modeled as primary): `high_human_impact` (death or injury), 272 positives, 1.7% prevalence.
- Leakage rules enforced (good rigor slide):
  - Never use outcome columns (deaths/injuries/damage) as predictors.
  - Split **chronologically** (train ≤2021, validate 2022–2023, test 2024–2026).
  - Group cross-validation folds by `episode_id` so one storm never spans train/test.
  - Missing spatial data stays missing — never imputed as zero.
  - Imperviousness is a static 2023 layer laid on top of older events — flagged as a modeling caveat.

PART D — Modeling: what you actually built

Slide: Three models, three trade-offs (beginner-friendly)

- **Logistic Regression** — one weighted sum per feature → probability. Fully explainable (every coefficient is a number you can show), but assumes each feature acts independently ("a straight line separating risky from safe").
- **Random Forest** — 100 decision trees, each trained on a random subset of data/features, votes averaged. Captures feature *combinations* a straight line can't, at the cost of being harder to explain per-prediction.
- **HistGradientBoosting** — also trees, but built sequentially, each new tree fixing the previous trees' mistakes. Often most accurate, least explainable of the three — reported for comparison, not deployed to either interactive tool.

Slide: Training recipe (the "we did this properly" slide)

1. **Leakage-safe labeling** — damage threshold computed pre-2022 only.
2. **Feature engineering** — real terrain/hydrology/infrastructure/land-cover features (see Part C), plus two engineered features added *after* testing:
   - flood history split into **minor** (builds resilience → lowers risk) vs. **severe** (compounds vulnerability → raises risk) — confirmed opposite-signed effects when tested; a genuine finding, not just an engineering trick.
   - `peak_hurricane_season` (Aug–Oct flag) — added after discovering the original `is_tropical_event` flag was **used in zero of the 100 trees**.
3. **Class weighting** (`class_weight="balanced"`) — only ~4–9% of events are high-consequence; without this the model just predicts "safe" every time.

4. **Grouped cross-validation** — same storm episode never split across train/val/test.
5. **Calibration** — Platt scaling fit on **out-of-fold** predictions, not in-sample ones (see bug story below).

Slide: Bug #1 — dead feature, mountain inversion

- Found via direct inspection: `is_tropical_event` was never split on by any of the 100 trees — the entire "hurricane vs. typical storm" distinction rode on raw `event_month` alone.
- Quantified the consequence: checked systematically across 90 real mountain grid cells — **82% showed a hurricane scenario as *less* risky than a typical storm**, the opposite of physical reality.
- Tried the fix (engineered `peak_hurricane_season` flag): improved overall accuracy (PR-AUC 0.220 vs. 0.208) but did **not** fully fix the inversion (85.6% of mountain cells still inverted) — disclosed as a structural limitation of only ~150 positive training examples, not a simple bug.
- The one guaranteed-correct fix: the **linear model** can't have this problem by construction (one fixed month coefficient applied identically everywhere) — stated directly as a disclaimer in the Random Forest tool.

Slide: Bug #2 — calibration (the "we caught a real bug" slide — judges like this)

- Original Random Forest calibration was fit on **in-bag** predictions (same rows used to train the trees) → artificially overconfident, capping the most extreme physically-maxed-out input at just **2.6%** predicted risk.
- Fix: refit calibration on **out-of-fold** predictions instead.
- Result: that same extreme-case ceiling roughly **tripled, 2.6% → 7.9%**.
- **Figure:** `calibration_plot.png` (reliability diagram, raw vs. calibrated side-by-side) — this is your single best "we found and fixed a real bug through testing" visual.

PART E — Results

Slide: Headline numbers

Report **both** numbers you have, labeled clearly as different protocols:

Repeated grouped 5×5 cross-validation (11-feature set, from `datathonM3.py`):

Model	PR-AUC	ROC-AUC
Logistic Regression	0.199 ± 0.062	0.730 ± 0.058
Random Forest	0.220 ± 0.097	0.738 ± 0.093
HistGradientBoosting	0.216 ± 0.075	0.714 ± 0.093
Base rate (random)	0.067	0.500
Published benchmark*	0.127	0.783

*Comparable Texas flash-flood study, similar features/rarity — your numbers beat it on PR-AUC.

- Talking point: PR-AUC (not ROC-AUC) is the metric that matters most here, because positives are rare (~4–9%) — a model can look good on ROC-AUC while being nearly useless at actually ranking the few true high-consequence events.
- **Figure:** `pr_curve.png` (single train/test holdout curve — label it clearly as illustrative alongside the CV table above, since it's a different, smaller-n evaluation).

Slide: What drives the predictions (explainability)

- SHAP summary plot for Random Forest: `event_month` / `peak_hurricane_season` dominate, then `impervious_mean_250m` (low imperviousness → higher risk, consistent with rural/mountain terrain being disproportionately dangerous), then terrain relief, elevation, and flood history.
- **Figure:** `shap_summary.png`.
- One dramatic single-event example (SHAP force plot) is a great "here's why the model flagged this one flood" slide if you have time to make it — see the SHAP steps from earlier in this chat.

Slide: Descriptive look before modeling (optional, if you want an EDA→ML bridge slide)

- `fig14_high_consequence_rate_map.png` / `fig15_features_by_target_descriptive.png` / `fig16_feature_correlation_matrix.png` — how features differ between high-consequence and other flood events, purely descriptive (label it "descriptive, not a model result" like the original report does).
- `fig17_statewide_flood_events_map.png` — where flood-related events became high-consequence, statewide.

PART F — Honest limitations (closing technical slide — keep this in, it reads as rigor not weakness)

- Point-and-buffer features only — no upstream contributing area, HAND, or flow accumulation; HUC12 membership isn't a real drainage-area measure.
- 4,279 events (mostly forecast-zone records) have no usable coordinates — excluded, never defaulted to a centroid.
- Imperviousness is a static 2023 layer applied to events across 2016–2026 — timing mismatch, worth checking how much the model leans on it.
- Rainfall and community-vulnerability data are not yet integrated into the spatial pipeline (SOVI/population were added at the modeling stage from FEMA NRI, but not into the full feature-engineering report above).
- `local_relief_5km` in the current model run is approximated from `roughness_mean_250m` pending the real 5km relief raster.
- Partial year 2026 is incomplete — don't compare it to full years.
- The mountain-inversion issue (Bug #1) is disclosed, not fully solved.

Suggested slide order (if you want one linear path)

1. Helene hook + core question
2. NOAA data source
3. Why NOAA alone isn't enough → 5 supporting datasets
4. Scale/region numbers + seasonality
5. Feature engineering methodology
6. Outcome label + leakage discipline
7. Three models explained
8. Training recipe
9. Bug #1 (dead feature / mountain inversion)
10. Bug #2 (calibration) + calibration plot
11. Results table + PR curve
12. SHAP / what drives predictions
13. Limitations